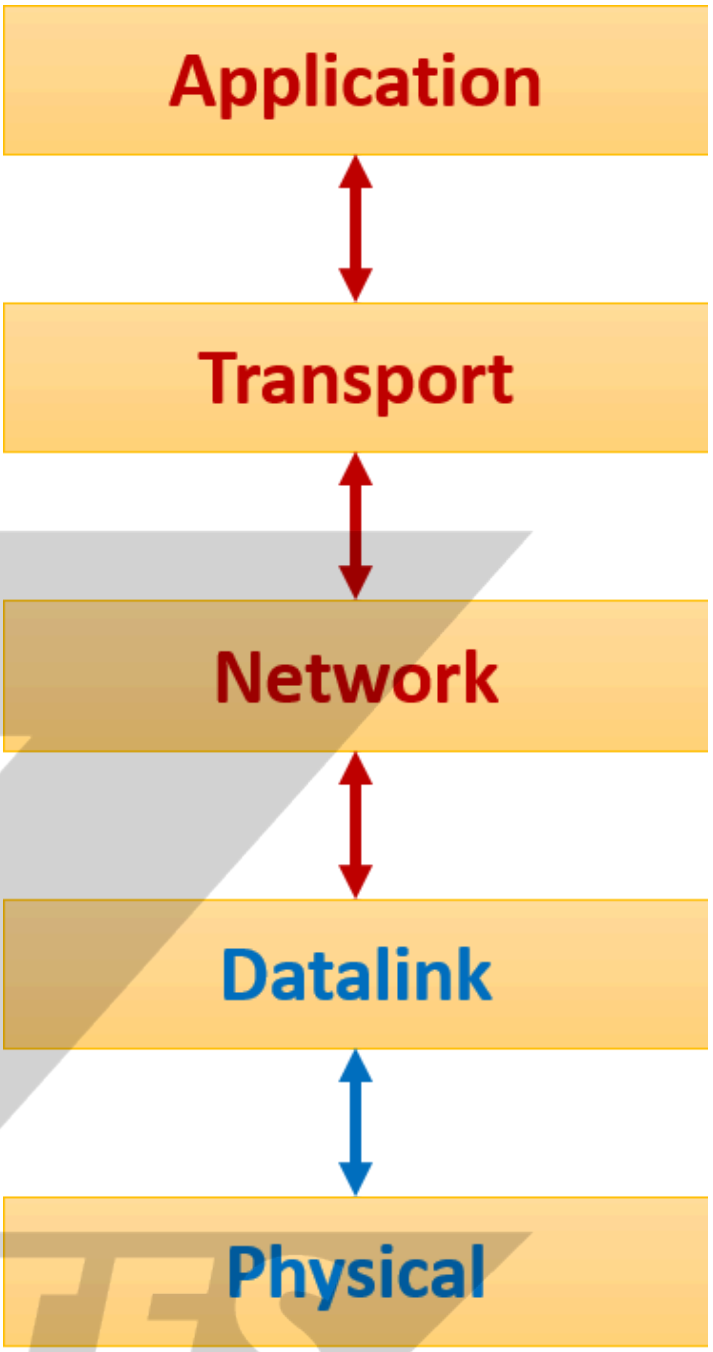
1. Communication and Internet Technologies

1.1. Protocols

A protocol sets the rules of the communication. Protocols allows for devices to communicate across different platforms. If two devices attempted to communicate while using different protocols, the communication would fail. TCP/IP is the dominant protocol suite for internet usage.

TCP/IP Suite

- A layered model (stack) with 4 layers; Application. transport, internet and data link. The Suite uses a set of protocols for the transmission of data in these layers like TCP (Transport Control Protocol) and IP (Internet Protocol).
- TCP maintains connection between two nodes and ensures delivery of data between the two nodes.
- IP provides rules of exchange for packets and decides the route for sending packets.



Application Layer:

- The application layer provides user services.
- Encodes data in an appropriate format when sending a message
- Performs the action requested by the user when receiving a message

There are the main types of protocols:

- HTTP and HTTPS (Hypertext Transfer Protocol): Used to transfer webpages from server to client. HTTPS is more secure.
- FTP (File Transfer Protocol): Used for sending and receiving files over the network between two devices.
- POP3 (Post Office Protocol): Used for receiving emails.

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- **IMAP (Internet Message Access Protocol):** Also used for receiving emails. The email client retrieves the emails from the mail server.
- **SMTP (Simple Mail Transfer Protocol):** Used for sending emails

Transport Layer:

- Handles packets.
- Converts the data received from the application layer into individual packets when sending a message.
- Adds packet headers and then sends these packets to the internet layer.
- Controls the flows of packets and handles packet loss or corruption.
- When receiving a message, it reads the header and determines which application layer the data must be sent to, then strips the header and forwards it

Internet Layer:

- Handles the transmission of data using IP addresses.
- Identifies the intended network and host and then transmits the packets through the data link layer.
- Routes each packet independently through the best route available
- IP Addresses, of the sender and receiver, and checksum are added when sending messages
- When receiving a message it reassembles the fragments, strips the IP header and sends it to the transport layer

(Data) Link Layer:

- Converts the electrical data into analog signals and sends it, allowing the upper layers access to the physical medium.
- Formats data into frames for transmission and maps IP addresses to MAC (hardware) addresses.
- Ensures correct network protocols are followed.
- Verifies the checksum when receiving a message and then sends it to the internet layer

Peer-to-peer File Sharing

- BitTorrent uses Peer-To-Peer file sharing model.
- A BitTorrent client software is made available to the public.
- **Swarm:** All connected peer computers that have all or parts of file being downloaded/uploaded.
- **Tracker:** A central server that stores information on which computers have which files and which parts of those files. It also shares the IP addresses of computers to allow them to connect to each other.
- **Seed:** is the peer computer that uploads the file that is being downloaded.

If a user wishes to join this network, they need to use a BitTorrent client to load the torrent descriptor file. When a user wishes to download a file, pieces of this file are downloaded and uploaded at the same time. When a user has even a single piece of a file they become a seed. To be able to download this file, a complete copy needs to exist on one of the peer computers.

1.2. Circuit Switching, Packet Switching

Circuit Switching

- Circuit Switching: Method of transmission in which a dedicated circuit lasts throughout the duration of the communication.
- Dedicated circuit is selected before the communication starts and is used to send all the data through this route. - Uses the whole bandwidth.
- Releases circuit once the communication ends.
- Used for live video broadcasts or calling
- Data arrives in order so it doesn't need to be reassembled.
- Whole bandwidth is available.
- Data can't get lost.
- Less secure as it only uses one route.
- No alternative route in case of failure.
- Bandwidth can't be shared.

Packet Switching

- Method of transmission where the message is broken down into packets which can be sent along independent paths
- Data is split into multiple packets and the best route is calculated for each packet.
- Each packet contain a header, which has a source IP address and destination IP address, and a payload.
- Calculates the most efficient path for each packet.
- Used when sending large files that don't need to be live streamed or when it is necessary to be able to overcome faulty lines by rerouting like emails, text messages, sending documents, etc.
- Harder to intercept as new route is used each time.
- Packets need to be reassembled.
- If there's a missing packet, a request is sent to retransmit the data.

Role of Router:

- Examines the header and finds the destination IP address.
- Decides the best route for the packet, with the help of the information from the routing table, and sends it towards the next hop.

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Comparison

	Circuit Switching	Packet Switching
Nature	Dedicated channel	Data segmented into packets
Setup	Requires pre-established channel	No prior setup needed
Data handling	Message remains intact	Message gets split into packets
Transmission	Single Path	Path recalculated each time
Order	In order	May need reassembly
Layer	(Data) Link Layer	Network Layer
Bandwidth	Makes full use of bandwidth	Bandwidth can be shared
Error handling	Communication ends incase of an error	Allows packets to be resent
Complexity	Simple	More complex

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